



- Problem Statement ID – **SIH26079**
- Problem Statement Title – **AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts**
- Theme – **Smart Automation**
- PS Category – **Software**
- Team ID – [48]
- Team Name (Registered on portal) – [**HexArk**]

VEYRA — Know When Forecasts May Fail

“Forecasts tell you what may happen. Veyra tells you when the forecast itself may fail.”

A student-built trust layer for the forecasts India relies on.



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

“Clear skies,” said the app. By noon the field was underwater — and nobody had warned that the forecast itself was shaky.

A BUST = forecast absolute error crossing the certified threshold

Story + photo (Plassey, WB — Pexels) are illustrative. Everything below it is built and tested.

Sowing day. One confident number on the screen — and a family's whole season riding on it. When forecasts fail, nobody warns you first. So we built the warning: Veyra, a second opinion on every forecast.

We don't predict weather. We grade the prediction.

Challenges and Problems

- Apps show one confident number and hide all doubt
- One bad call can wipe out a sowing, a flight slot, a harvest
- No product today tells you: “don't trust me this time”
- So people either trust blindly — or stop trusting at all

Proposed Solution

- **A second opinion on every forecast** — sits on NOAA GEFS ensembles; scores the odds of a bust before the event.
- **Safety-first answers** — a risk band with plain-English reasons, or a straight refusal. We never guess.
- **Built to fit in** — a trust layer inside IMD-style dissemination, not a rival forecast.

Innovation and Uniqueness

- **We grade predictions, not weather** — rides along any NWP model, never competes.
- **Anti-Leakage by Design** — tests prevent future information from entering the live path.
- **Calibrated Risk + Abstention** — calibrated risk with reasons, or an abstained response.

How it addresses the problem:-- HOW IT ADDRESSES THE PROBLEMS | | Hidden uncertainty → Reliability signal | | Forecast fragility → Trajectory fingerprint | | False confidence → Calibration + abstention | | Generic forecast → Place-specific risk .

TECHNICAL APPROACH

❖ Technologies stack • Methodology and process for implementation

Methodology and Process

System Architecture (working Prototype)

Technology Stack



• Python • FastAPI



• React • TS



• sklearn • Pydantic

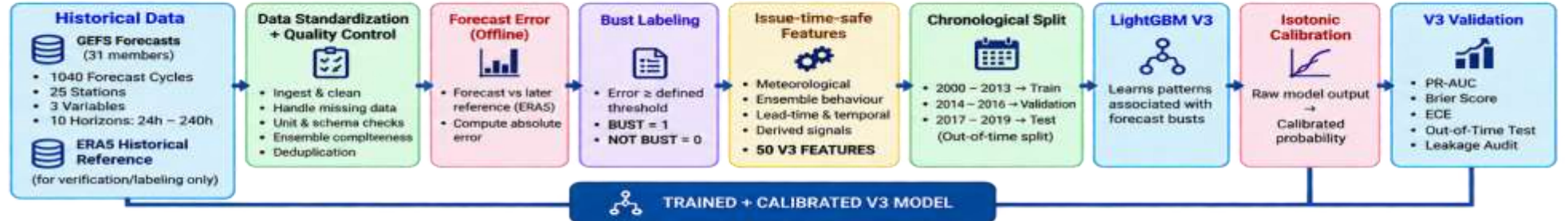


• Vite • pandas

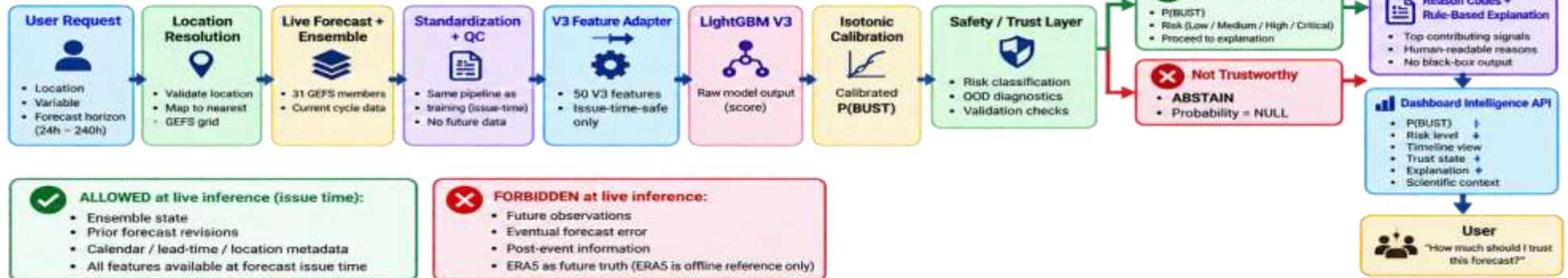
✓ **ALLOWED at issue time:**
ensemble state • prior
revisions • calendar codes

System Architecture (working Prototype)

OFFLINE TRAINING & VALIDATION (Model Development)



LIVE INFERENCE PATH (Operational Use)



• **LIVE PROTOTYPE OUTPUTS :** Kolkata P(bust) = LOW New Delhi = LOW CALIBRATED
VEYRA V3 • LIGHTGBM + ISOTONIC CALIBRATION

Live inference uses only issue-time-available information

❖ Analysis of feasibility • Challenges & risks • Strategies

CURRENT FEASIBILITY

1. Technical

Current Prototype : Fast API + React/TypeScript, end-to-end working workflow, automated tests, modular architecture.

2. Data & Cost

Public-proxy feasible: GEFS + ERA5 public research data, low-cost prototype path, pipeline designed for extension.

3. Operational

Research prototype verified end-to-end; operational deployment requires institutional validation and deployment infrastructure.

4. Evidence

Evidence: frozen out-of-time benchmark + calibration + leakage audit.

$PR-AUC \approx 0.211 \cdot Brier = 0.0538 \cdot ECE = 0.0068 \cdot Brier Skill Score = +0.077$

What could go wrong → What we do

Rare bust events

→ Recall-oriented evaluation, calibrated risk assessment, abstention

Data / API wobbles

→ Caching, request deduplication, input validation, adapter-based source isolation

Institutional validation gap

→ Paired historical data, metadata mapping/harmonization, independent verification

FEASIBILITY BOUNDARY — ADVANCEMENT ROADMAP

CURRENT
PUBLIC-PROXY
PROTOTYPE

verified prototype workflow

ROBUST
DATA ENGINE

quality gates · provenance · schema
validation · harmonization

ADVANCED
RELIABILITY
INTELLIGENCE

run-to-run divergence · regime
conditioning · analog trajectories · spatial
risk maps

NCMRWF/NEPS
CONDITIONAL
VALIDATION

*FUTURE/CONDITIONAL: paired data,
metadata mapping, permission/access,
independent verification*

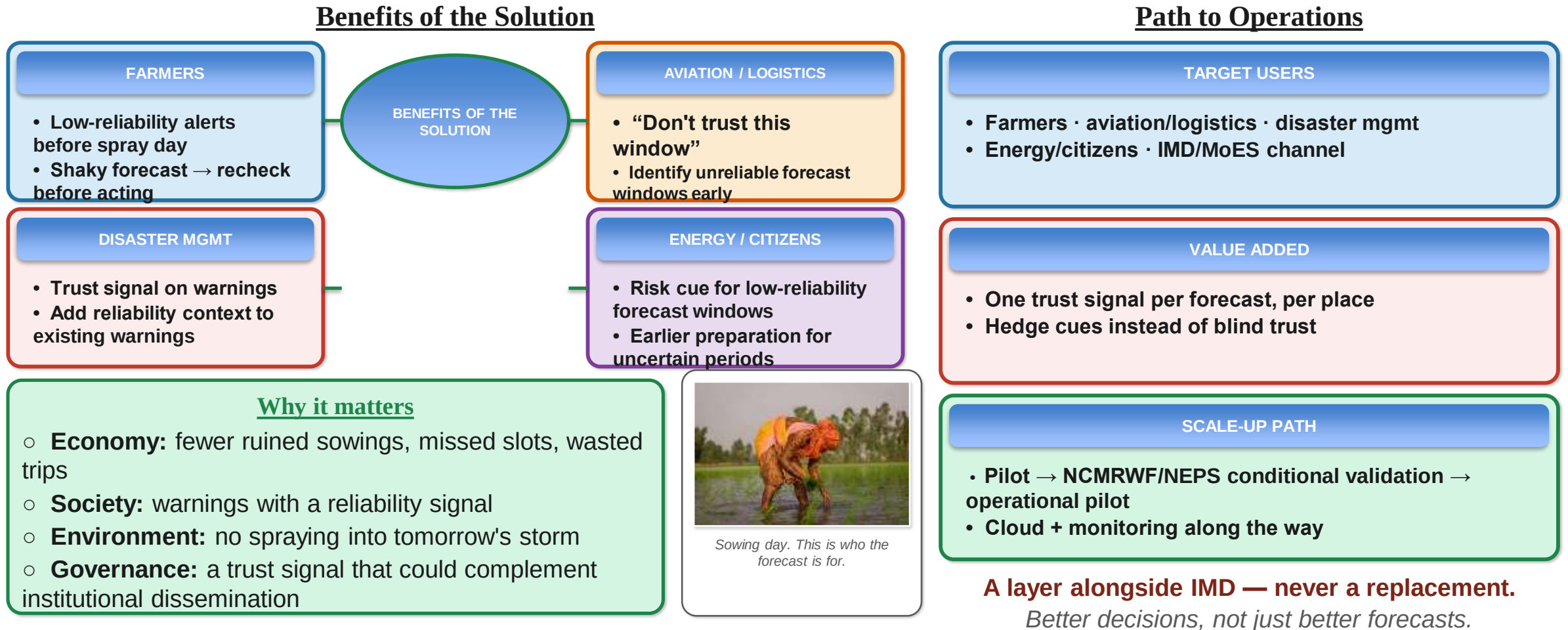
INDEPENDENT
RESEARCH PILOT

future milestone, not current status

“Feasible today within the public-proxy research scope. Operational deployment requires additional data access, validation and independent verification.”

We distinguish what is built, what is verified, and what remains to be proven.

❖ Potential impact on the target audience • Benefits of the solution



Details / Links of the reference and research work

1] NOAA/NCEP — Global Ensemble Forecast System (GEFS).

Primary ensemble forecast source; GEFS provides multiple perturbed forecasts to represent forecast uncertainty.

Link: [NOAA GEFS Official Source](#)

[2] ECMWF — ERA5 Reanalysis.

Historical reference/reanalysis dataset used for offline verification and analysis; **not used as future truth in live inference.**

Link: [ECMWF ERA5 Official Source](#)

[3] Rasp et al. — WeatherBench 2 (2023).

Benchmark and evaluation framework for comparing data-driven and traditional global weather forecasting methods.

Link: [WeatherBench 2 Repository](#)

[4] Open-Meteo — Weather API.

Public weather-data access layer used for prototype integration and data retrieval.

Link: [Open-Meteo Official API](#)

[5] Isotonic calibration reference — use the verified source from the V3 research record.

Placeholder — no citation invented; V3 uses isotonic calibration, not Platt scaling.

Agar space bache toh:

[6] Forecast Verification & Ensemble Predictability Literature.

Used as methodological background for forecast uncertainty, verification and ensemble-based reliability analysis.

[7] Project Research Archive — Veyra.

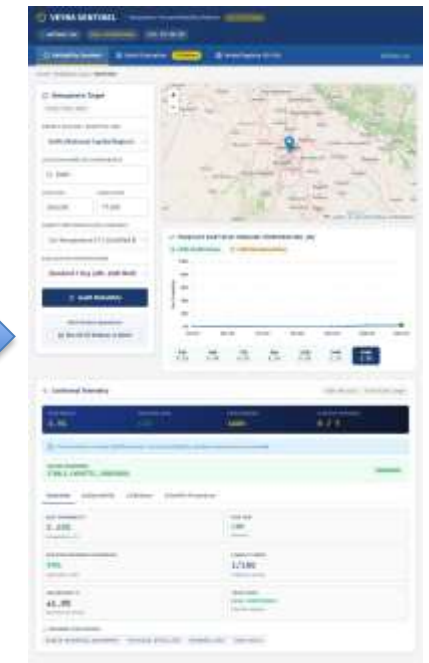
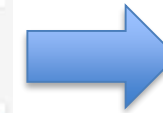
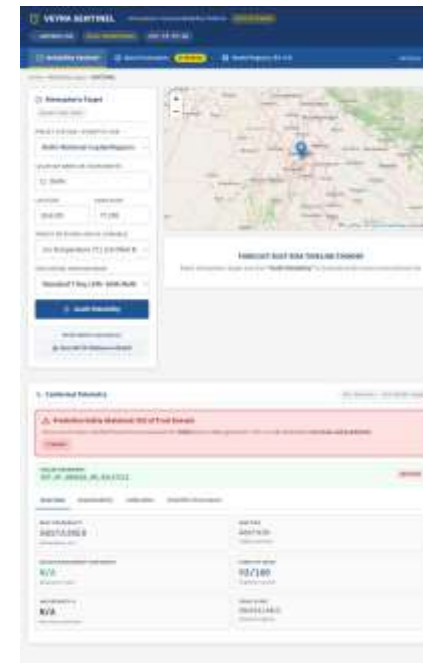
Internal research documents covering canonical harmonization, issue-time safety, trajectory instability, calibration, evaluation and validation boundaries.

TEAM DEMO / PROJECT PORTAL

don't take our word for it

Veyra — Know When Forecasts May Fail

<https://adishxm.github.io/Veyra-Know-When-Forecasts-May-Fail/>



Sample UI by the team (real screenshot): *safe abstention in action* — refuses to guess outside its trust domain.

CURRENT: PUBLIC-PROXY RESEARCH PROTOTYPE | FUTURE: NCMRWF / NEPS
CONDITIONAL VALIDATION